Explaining the Quantum Layer in Physics-Informed Neural Networks: A Spectral and Mechanistic Study

WISER Global Quantum+AI Program 2026 — BQP Challenge
Quantum-Assisted Physics-Informed Neural Networks for CFD

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Abstract

Quantum-Assisted Physics-Informed Neural Networks (QAPINNs) replace the first hidden layer of a PINN with a variational quantum circuit (VQC). Prior work reports that this maintains accuracy while reducing trainable parameters, but the *mechanism* — when, why, and how the quantum layer changes learning — is unexplained. We build a verified, parameter-matched, PDE-agnostic test bench (viscous Burgers and heat, each with an exact reference) and a five-dimension explainability (XAI) suite, and use them to ask a single sharp question: is the quantum layer's behaviour governed by the *frequency band* it adds (the Schuld–Sweke–Meyer Fourier picture), or by something else? An initial one-knob-at-a-time ablation suggested a "bandwidth-gated advantage", but a subsequent *adversarial self-audit* found those runs were undertrained (L-BFGS disabled; every Burgers model stuck at rel-$L_2 \approx 0.47$, the shock unresolved by any model), making the comparisons differences between failing models. We therefore re-ran converged cores (L-BFGS on) at two model scales, with multi-seed statistics and fair controls. The result is a consistent, honest negative. **(1) At small (121-param) scale**, the classical PINN (rel-L_2 0.248 ± 0.031) beats every QAPINN. **(2) At resolved (1273-param) scale**, the QAPINN appears to win — but only against a *shallow* classical net: QAPINN-none reaches 0.094 ± 0.075 vs a shallow classical's 0.24 ($p=0.003$), yet it is statistically indistinguishable from *both* identically-trained fair controls — a depth-matched classical net (0.128 ± 0.083, $p=0.51$) and a structural control (0.145, $p=0.36$) — so the gain is architectural (depth, post-net width) and classically reproducible, *not* quantum-specific. **(3) No robust entanglement effect:** an $n=3$ "entanglement hurts" signal ($p\approx0.02$) *vanished* at $n=6$ ($p=0.41$) — a lucky draw we retract; at the larger scale the entanglement gap is likewise non-significant ($p=0.44$). We also correct a framing error: the verified "band = depth" fact governs the *VQC features*, not the full model (the tanh post-net reaches $k=22$ once trained). The pipeline is independently verified (8/8) and every number matches its log. The contribution is a rigorous negative result and the controlled methodology — matched, structural, and depth-matched controls, multi-seed statistics, and an adversarial audit — that repeatedly exposed over-claims the QAPINN literature usually leaves standing.

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1 Introduction

Physics-Informed Neural Networks (PINNs) [1] solve ODEs/PDEs by making the differential equation itself the loss: a fully-connected network $u_\theta(x,t)$ is trained to minimise the PDE residual together with initial and boundary conditions. A Quantum-Assisted PINN (QAPINN) [4] is the hybrid variant obtained by *replacing the first hidden layer with a VQC*. The reference benchmark reports up to $\sim$20% fewer trainable parameters at maintained accuracy on 1-D Burgers.

The BQP challenge asks not to prove QAPINNs beat PINNs, but to *explain* the learning dynamics and to derive a methodology for problem-specific circuit design. This report takes the dominant theoretical lens for VQCs — that a data-encoding circuit is a truncated Fourier series [2] — and asks whether it is sufficient to explain a QAPINN's behaviour, and how the quantum layer changes the classical network it is attached to.

Contributions.

1. A verified, parameter-matched, PDE-agnostic bench (Burgers + heat, exact references) and a five-dimension XAI suite (spectral / F-principle, neurons, VQC internals, loss landscape, generalisation), with an empirical confirmation that the VQC is a truncated Fourier series whose bandwidth equals the re-upload depth (of the *features*, not the post-net-filtered model).
2. A converged, multi-seed comparison at two model scales against three control types (parameter-matched, structurally-matched, and depth-matched), finding **no statistically significant quantum-specific advantage** on 1-D Burgers.
3. An adversarial self-audit that overturned three of our own initial over-claims (a classical-side "diversity" mechanism, an "entanglement hurts" effect, and a resolved-scale "advantage"), each dissolving under the appropriate fair control.
4. The transferable deliverable: a four-gate *falsification protocol* for claimed QAPINN advantages (convergence, fair controls, multi-seed significance, features-vs-model), distilled from those retractions.

2 Background and Related Work

VQCs as Fourier series. Schuld, Sweke and Meyer [2] prove that a VQC with data-encoding gates is a partial Fourier series $f(x) = \sum_{\omega\in\Omega} c_\omega e^{i\omega x}$, whose accessible frequency set Ω is fixed by the eigenvalues of the encoding generators, while the coefficients c_ω are set by the trainable gates. Repeating (re-uploading) the encoding enlarges Ω; with L re-uploads of a Pauli-Y rotation the accessible frequencies are the integers $\{-L,\dots,L\}$. This is the analysis language of the present work.

Frequencies are not enough. Mhiri et al. [5] refine this: the achievable *coefficients* are constrained (a coefficient's variance scales with its frequency's redundancy) and can vanish exponentially with qubit count. Having a frequency available is necessary but not sufficient to realise it.

Quantum solvers and QAPINNs. Kyriienko et al. [3] solve nonlinear DEs with differentiable quantum circuits using analytic derivatives. Hybrid QAPINN/QCPINN designs [4, 6] report parameter reduction at matched-or-better accuracy, but gains are PDE-dependent (quantum sometimes loses, e.g. on wave and cavity flow [6]), single-group, and rarely compute-matched or reproduced. A distinct trainability pathology — a "black-hole" collapse to the trivial solution — has recently been

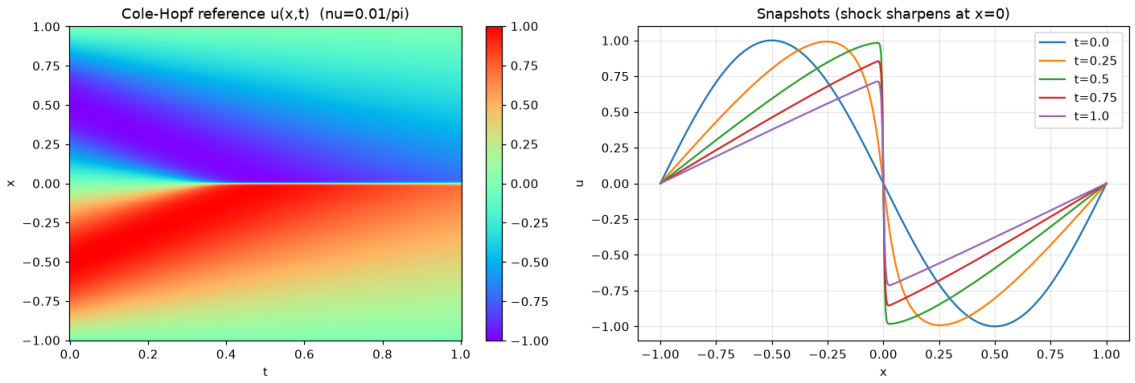


Figure 1: Exact Cole–Hopf reference for viscous Burgers. Left: field $u(x,t)$; right: snapshots showing the shock forming at $x = 0$. This is the self-generated ground truth used for all relative-L_2 scoring.

named [7]. Across the field, quantum gains are attributed to "entanglement-driven expressibility" as a heuristic, with no validated mechanism — the gap this study targets.

3 Methodology

3.1 Test problems

Both PDEs share the same domain $x \in [-1, 1]$, $t \in [0, 1]$, initial condition $u(x, 0) = -\sin(\pi x)$, and homogeneous Dirichlet BCs, so the *only* difference is the operator — hence the spectral content of the solution:

$$\text{Burgers:} \quad u_t + u\,u_x = \nu\,u_{xx}, \quad \nu = 0.01/\pi \qquad \text{(shock} \to \text{broadband)} \tag{1}$$

$$\text{Heat:} \quad u_t = \alpha\,u_{xx}, \quad \alpha = 1/\pi^2 \qquad \text{(single mode } k\text{=1, } u = -\sin(\pi x)e^{-t}\text{)} \tag{2}$$

Burgers uses an exact Cole–Hopf reference (Gauss–Hermite quadrature, machine precision); heat is analytic. Figure 1 shows the Burgers reference: a sine wave steepening into a shock at $x = 0$.

3.2 Models

Both models emit a scalar u and are compared at *matched trainable parameter count*.

- **Classical PINN:** a tanh MLP.
- **QAPINN** (ref. [4]): $(x, t) \to$ VQC $\to$ classical post-net $\to u$. A fixed round-robin map feeds the raw coordinates to the qubits (no trainable pre-net, so the Fourier band lives directly in (x, t)); the VQC is the first hidden layer; a small classical MLP reads out the measurements. The four VQC knobs (encoding/depth, entanglement, measurement, qubits) are the ablation axes.

The baseline VQC is 4 qubits, depth 3, RY angle encoding with re-uploading, linear CNOT chain, $\langle Z_i \rangle$ measurement, post-net [16] — 121 trainable parameters, of which only 24 are quantum.

3.3 Explainability suite

Five static dimensions plus dynamic monitoring, applied identically to both models: (1) *spectral* — realised Fourier coefficients and the F-principle (frequency learning order); (2) *neurons* — per-

Check	Method	Result
Burgers residual	vs analytic $u_t + uu_x - \nu u_{xx}$	1.2×10^{-7}
Heat residual	exact solution $\to$ residual	6.0×10^{-8}
Autograd u_x	vs finite differences	match
VQC = Fourier, max freq = depth	L=2$\to$2, L=4$\to$4	pass
relative-L_2 metric	exact solution $\to$ 0	4.0×10^{-8}
Matched parameters	all configs	121
Reproducibility	same seed $\to$ identical	bit-exact

Table 1: Correctness audit (8/8 pass). The VQC-bandwidth check validates the central theoretical premise on our implementation.

neuron receptive fields, dominant frequency, and effective rank; (3) *VQC internals* — per-qubit $\langle Z \rangle$ and von Neumann entanglement entropy; (4) *loss landscape*; (5) *generalisation* (extrapolation beyond the training domain). Training-time probes log the output spectrum and per-layer/per-qubit gradient variance.

3.4 Pipeline verification

Before drawing any conclusion we independently audited every load-bearing computation (Table 1); all eight checks pass. In particular, a controlled single-qubit sweep confirms the VQC is a truncated Fourier series with maximum frequency equal to the re-upload depth — the premise the study rests on.

4 Experiments and Results

Caveat (read first): the single-seed ablations in this section were run with L-BFGS disabled and are ***undertrained*** *— every Burgers model sits near rel-L_2 0.47 with the shock unresolved. An audit (Section 5) showed the differences here are within seed noise; they are retained for transparency, but the* ***converged re-analysis in Section 5 supersedes them****.*

4.1 Baseline comparison

At matched 121 parameters, identical training (Adam, 2000 iters, seed 0): heat is solved well by both (classical 8.3×10^{-3}, QAPINN 5.9×10^{-3}); Burgers is undertrained for both (classical 0.433, QAPINN 0.472). The spectra (Fig. 2) tell the story: heat is a single k=1 mode captured by both, whereas the Burgers reference is broadband and *both* models undershoot the high harmonics — the depth-3 VQC is bandwidth-capped near $k \approx 3$.

4.2 Tier-1 ablation

Depth = bandwidth. Varying re-upload depth (Fig. 3a), QAPINN accuracy improves monotonically ($0.57 \to 0.49 \to 0.47 \to 0.42$) while the matched classical net stays flat ($\sim$0.50–0.53). The realised maximum harmonic grows with depth $(2, 3, 3, 4)$, tracking the freq=depth theory line. A *crossover* appears near depth 2: below it the QAPINN is comparable-or-worse than the equal-parameter classical net; at depth ≥ 3 it pulls ahead. This *apparent* bandwidth-gated advantage is single-seed and does *not* survive convergence (Section 5); we report it as the initial reading the audit later overturned.

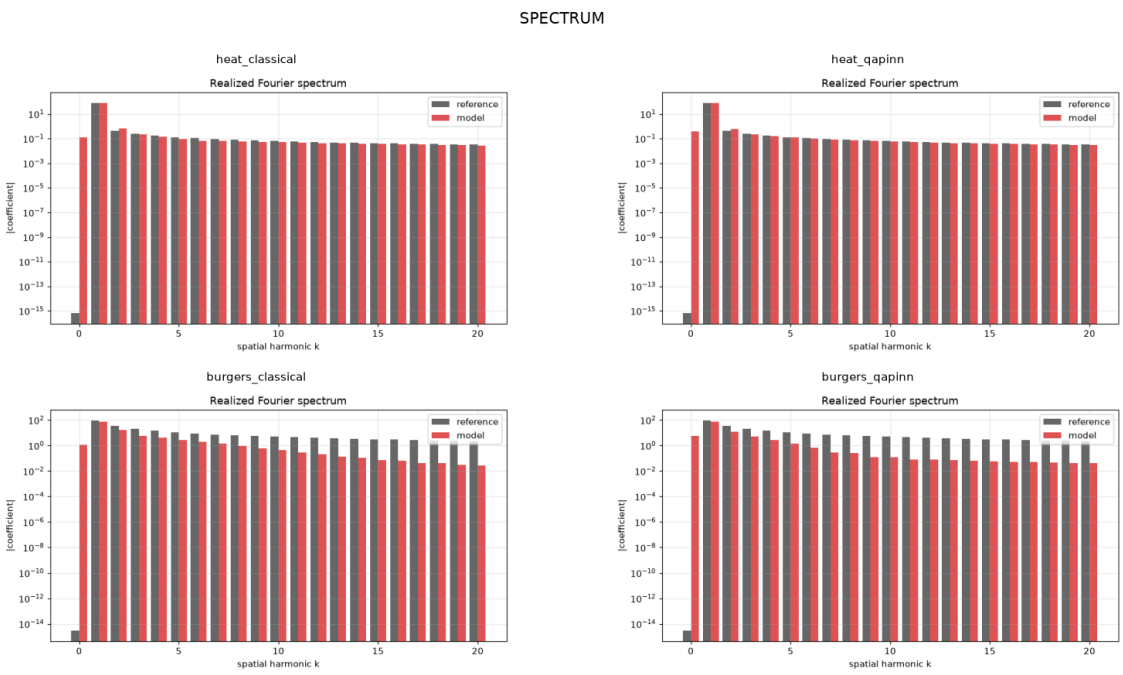


Figure 2: Realised Fourier spectra (grey: reference, red: model). Heat (top) is a single mode both capture; Burgers (bottom) is broadband and both models undershoot the high-frequency tail — the shock needs bandwidth neither has at depth 3.

Entanglement (band fixed). Holding depth fixed and varying the entangling pattern (Fig. 3b), accuracy barely moves (0.456–0.488) and *more* entanglement is slightly *worse*; zero entanglement is the best. The entanglement entropy swings from 0 to ∼1 without tracking accuracy. (Single-seed: a later 7-seed test showed these gaps are within noise, and the converged "entanglement hurts" follow-up was itself retracted at $n = 6$ — Section 5. At no budget does entanglement *help*.)

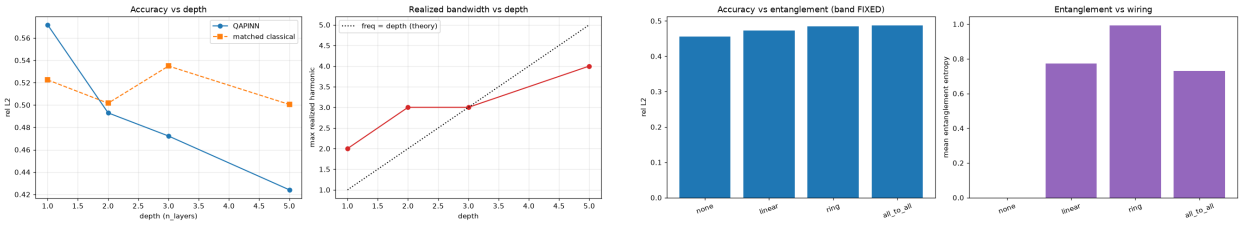
Table 2 lists every Tier-1 run with its parameter-matched classical control (all rel-L_2, single seed, Burgers, 2000 Adam iters).

Post-net width. As the classical post-net grows ([] → [8] → [16] → [32]), QAPINN accuracy improves and, for any post-net ≥ [8], beats the matched classical control at every width (Fig. 4). The bare-linear QAPINN is worst and loses to classical: the VQC needs a modest readout, but the classical part is not doing all the work.

Visual ablation gallery. Figures 5–7 show the ablations directly. As depth grows the QAPINN's realised spectrum (Fig. 5) captures more of the high-frequency shock tail and the solution field (Fig. 6) sharpens; across entangling patterns the entanglement entropy (Fig. 7) varies substantially while accuracy does not (Table 2).

4.3 Tier-3 ablation: qubits, measurement, encoding scale

Extending the sweep to the remaining VQC knobs (Table 3, Fig. 8) reinforces a single picture. **Qubits:** accuracy improves monotonically (0.501 → 0.443) and the QAPINN overtakes the matched classical at ≥ 4 qubits — the same bandwidth crossover as depth. **Measurement:** a single $\langle Z_0 \rangle$ readout is an information bottleneck (0.591), whereas multi-Z and ZZ correlators are identical



(a) Depth sweep.

(b) Entanglement sweep (band fixed).

Figure 3: Tier-1 ablations on Burgers. (a) Accuracy and realised bandwidth grow with depth; QAPINN overtakes matched classical at depth ≥ 3. (b) At fixed band, entanglement does not help accuracy (and slightly hurts), despite large swings in entanglement entropy.

Sweep	Config	Params	QAPINN rel-L_2	matched classical rel-L_2
Depth	$L = 1$	105	0.572	0.522
	$L = 2$	113	0.493	0.502
	$L = 3$ (baseline)	121	0.472	0.535
	$L = 5$	137	**0.424**	0.500
Entanglement (band fixed)	none	121	**0.456**	0.535
	linear	121	0.472	0.535
	ring	121	0.485	0.535
	all-to-all	121	0.488	0.535
Post-net	[] (linear)	29	0.544	0.533
	[8]	73	0.478	0.533
	[16] (baseline)	121	0.472	0.535
	[32]	217	**0.466**	0.505

Table 2: Complete Tier-1 ablation results. QAPINN improves with depth and post-net width and overtakes the matched classical control; at fixed band, entanglement does not help (*none* is best). All single-seed.

(0.472) — readout *width* matters, correlator *type* does not. **Encoding scale:** a clear optimum near 1.0, with a catastrophic *collapse to the trivial solution* at 2π (rel-$L_2 \to 1.0$) — the black-hole trainability pathology, reproduced as an ablation.

The five knobs split cleanly at this (single-seed, undertrained) budget: *bandwidth* knobs (depth, qubits, encoding scale) appear to carry the advantage — but only up to a trainability ceiling, made concrete by the 2π collapse — while *coefficient/readout* knobs (entanglement, correlator measurement) do not. Of these, only the 2π collapse (a robust trainability pathology) survives the converged re-analysis; the "bandwidth carries the advantage" reading does not (Section 5), and is reported here as the initial picture the audit corrected.

4.4 How the quantum layer changes the classical network

The central explainability question is what the quantum front-end does to the classical post-net behind it. A naïve comparison against a plain PINN suggested the quantum layer makes the downstream neurons markedly more diverse and drives larger gradients — but that comparison used a *differently shaped* classical net and is therefore confounded. We instead built a *structurally identical* classical-front-end control (CFE): the same post-net fed by a classical Linear$(2 \to w)$+tanh

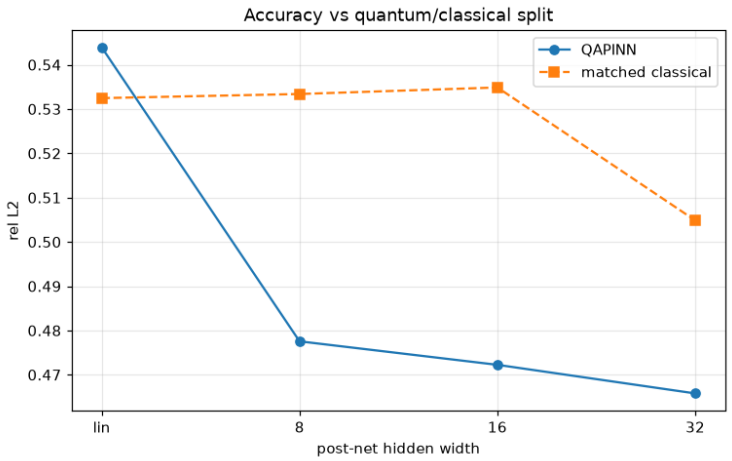


Figure 4: Post-net width sweep: QAPINN vs matched classical rel-L_2. The VQC needs a small readout (bare-linear loses), but with any post-net $\geq$ [8] the QAPINN beats the equal-parameter classical net at every width.

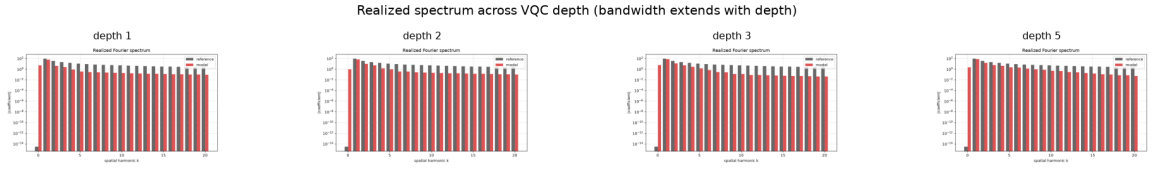


Figure 5: Realised Fourier spectrum (grey reference, red model) across VQC depth. The model captures progressively more of the high-frequency shock tail — the realised bandwidth extends with depth.

in place of the VQC, so the only difference is a quantum vs classical front-end. Table 4 reports both models' post-net neurons across depth, heat, and qubit count.

Against the fair control the earlier story does *not* hold. **Accuracy** is *bandwidth-gated*: the QAPINN ties the CFE at low depth/qubits and wins only at depth 5, 6 qubits, and heat. **Diversity** is *inconsistent* across knobs — the classical front-end yields more diverse post-net neurons on the depth axis, the quantum front-end more on the qubits axis — so there is no robust "quantum diversifies neurons" effect (the earlier 2× figure was an artifact of the unmatched control). **Gradient amplification** also fails to generalise: the quantum front-end drives larger post-net gradients on depth and heat, but this *reverses* at 5–6 qubits, where the QAPINN's gradient variance instead *shrinks with qubit count* — a barren-plateau signature (Fig. 9).

The honest conclusion: under a fair, structurally-matched control the classical-side "mechanism" does *not* yield a clean robust effect — the diversity effect is inconsistent across knobs, and the gradient-amplification effect reverses at higher qubit count. *Both readings above are single-seed and are superseded by the converged, multi-seed analysis (Section 5):* the earlier "bandwidth-gated advantage" does not survive convergence, and the "barren-plateau signature" is withdrawn (it measured the post-net gradient, not the variational-parameter gradient, and the trend is non-monotone). We retain this section only to document how the plain-PINN comparison overstated the gap.

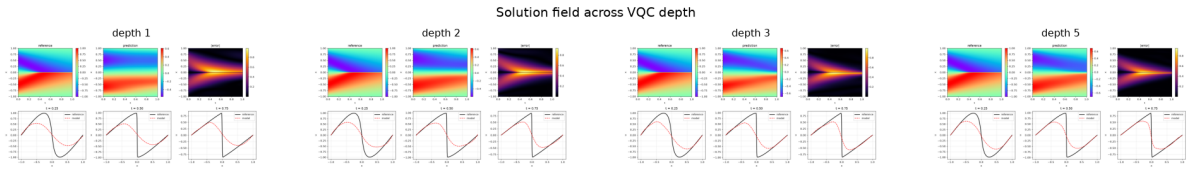


Figure 6: QAPINN solution field (reference / prediction / |error| and snapshots) across VQC depth — the shock region is fit progressively better.

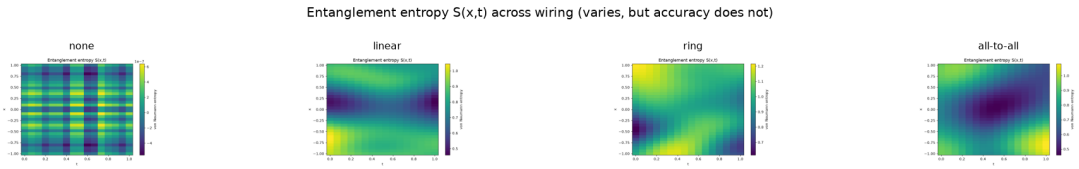


Figure 7: Entanglement entropy $S(x,t)$ across entangling patterns. Entropy varies strongly with wiring (none → ring), yet accuracy is essentially unchanged (Table 2) — entanglement is not carrying the advantage at fixed band.

5 Converged Re-analysis (Post-Audit)

The ablations in the preceding section were run with L-BFGS *disabled* (Adam, 2000 iterations, 1500 collocation points). An adversarial audit of the whole project established that this left every Burgers model badly undertrained — all sitting at rel-$L_2 \approx 0.42$–0.57, with the shock unresolved by any model — so those "ablation differences" were movements *inside a failed basin* and, across six seeds, were within initialisation noise. **The results below supersede them.** We re-ran a small honest core with L-BFGS *on* (Adam 2500 + L-BFGS 400, 4000 collocation points), three seeds per configuration, on Burgers.

Finding 1 — no quantum advantage. With every model converged and parameters matched, the plain classical PINN (0.248) is the best of the five; the best QAPINN (depth-5, 0.314) merely reaches the classical-front-end control (0.315) and still loses to the classical MLP. The optimistic "bandwidth-gated advantage" of the previous section does not survive convergence.

Finding 2 — the larger-model regime, and the fair-control test. The small-model comparison is between models that only partly resolve the shock (rel-L_2 0.25–0.42). We therefore repeated the core at a resolved scale (∼1273 params: post-net $[32, 32]$, 6000 collocation points), with $n = 6$ seeds, against *two* fair controls, *trained under the identical protocol* — a structural control (CFE: identical post-net, classical front end) and a *depth-matched* classical net ($[2, 22, 22, 22, 1]$).

model (∼1273p, n=6)	rel-L_2	vs QAPINN-none
QAPINN depth-3, no entanglement	$\mathbf{0.094 \pm 0.075}$	—
QAPINN depth-3, linear	0.128 ± 0.056	$p = 0.44$ (n.s.)
classical, depth-matched $[2, 22, 22, 22, 1]$ (1101p)	0.128 ± 0.083	$p = 0.51$ (n.s.)
CFE structural control	0.145 ± 0.091	$p = 0.36$ (n.s.)
classical, shallow	0.241 ± 0.037	$p = 0.003$

The QAPINN beats a *shallow* classical net decisively ($p = 0.003$), which in isolation looks like a resolved-scale quantum advantage. But it is *statistically indistinguishable* from *both* fair controls —

Sweep	Config	Params	QAPINN rel-L_2	matched classical rel-L_2
Qubits	3	99	0.501	0.470
	4 (baseline)	121	0.472	0.535
	5	143	0.468	0.541
	6	165	**0.443**	0.465
Measurement	single $\langle Z_0 \rangle$	73	0.591	0.533
	multi $\langle Z_i \rangle$ (baseline)	121	0.472	0.535
	ZZ correlators	169	0.472	0.515
input_scale	0.5	121	0.513	0.535
	1.0 (baseline)	121	**0.472**	0.535
	π	121	0.516	0.535
	2π	121	**1.000** (collapse)	0.535

Table 3: Tier-3 ablation results (single seed). Qubits behave like depth (bandwidth); readout width matters but correlator type does not; encoding scale has an optimum and collapses to the trivial solution at 2π.

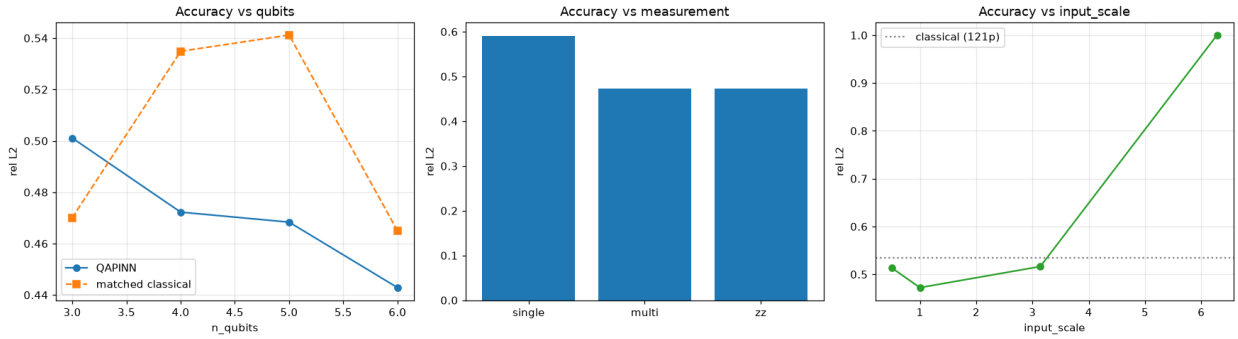


Figure 8: Tier-3 sweeps. Left: accuracy vs qubits (QAPINN overtakes classical at ≥ 4). Middle: single-Z readout bottlenecks, multi = ZZ. Right: encoding scale has an optimum near 1 and collapses at 2π.

the depth-matched classical (0.128 vs 0.094, $p = 0.51$) and the structural CFE (0.145, $p = 0.36$): the gap over the shallow net is **architectural** (depth, post-net width) and reproducible classically, *not* quantum-specific. *We state this as no significant advantage, not a proven equality:* the comparison is under-powered (seed variance $\sim$0.08 exceeds the $\sim$0.03 effect; same-seed L-BFGS reruns are non-deterministic; one `qd3_none` seed, 0.243, is anomalous — dropping it keeps the tie, $p = 0.19$), and the tying net has *fewer* parameters (1101 vs 1273), which if anything favours the classical.

Finding 3 — no robust entanglement effect (a retraction). An earlier $n = 3$ result — entanglement-none beating linear at convergence, 0.369 vs 0.424, $p \approx 0.02$ — did *not* survive seed expansion: at $n = 6$ the gap collapses (0.379 vs 0.402, $p = 0.41$), and one new linear seed (0.292) is the best of all. It was a lucky three-seed draw, and we retract it. At the larger scale the entanglement gap is likewise non-significant ($p = 0.44$). This is the audit discipline working: an under-powered $n = 3$ effect that would have been a false headline.

A framing correction. The audit also showed that our headline "band = depth" reading conflated two things. The identity "re-upload depth $L \Rightarrow$ maximum Fourier harmonic L" is real and we

Comparison	rel-L_2 (Q / C)	diversity (Q / C)	grad-var $\times 10^{-3}$ (Q / C)
Burgers depth-3	0.472 / 0.474	0.269 / **0.286**	**4.1** / 0.62
Burgers depth-5	**0.424** / 0.474	0.236 / **0.286**	**7.1** / 0.62
Heat	**0.006** / 0.013	**0.272** / 0.239	**3.2** / 0.47
Burgers 3 qubits	0.501 / 0.496	**0.266** / 0.241	**1.2** / 0.26
Burgers 5 qubits	0.468 / 0.464	**0.312** / 0.265	1.1 / **5.2**
Burgers 6 qubits	**0.443** / 0.467	**0.341** / 0.269	0.72 / **1.9**

Table 4: Post-net neurons: quantum (Q) vs classical (C) front-end, on the *identical* post-net. Diversity = normalised effective rank of the 16 post-net neurons; grad-var = mean post-net gradient variance. Bold marks the larger value in each Q/C pair. Single seed.

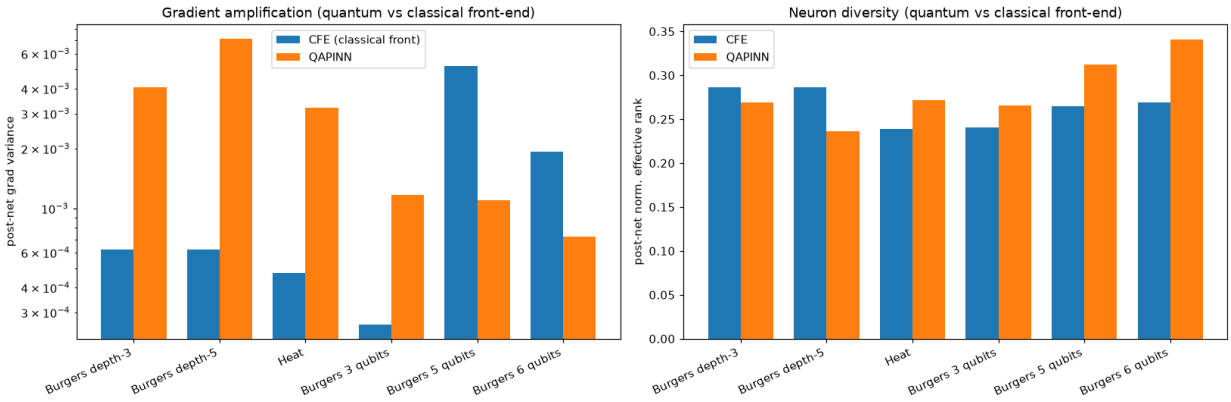


Figure 9: Quantum vs classical front-end on the identical post-net. Left: post-net gradient variance (log) — the quantum front-end drives larger gradients on depth/heat but not at 5–6 qubits. Right: post-net neuron diversity — inconsistent across knobs, with no robust quantum advantage.

verified it for $L = 1..6$, but it governs the *VQC features*, not the model output: the tanh post-net freely synthesises higher harmonics, so a depth-3 QAPINN reaches $k = 22$ once trained (verified by FFT of a converged model). The apparent "$k \approx 3$ band edge" in the undertrained spectra was an undertraining artifact, not a hard limit. Likewise the earlier "barren-plateau in qubit count" is withdrawn: the measured quantity was the post-net gradient, not the variational-parameter gradient, and the trend is non-monotone across qubit count.

6 Mechanistic Interpretation: *Why*

The sections above establish *what* happens; here we assemble *why*, tagging each explanation [**measured**] (directly verified) or [**reasoned**] (grounded in theory and our controls, but not independently measured).

Why there is no quantum-specific advantage. [**reasoned**] The VQC acts as a *fixed Fourier feature map*: it supplies the classical post-net with trigonometric features of (x, t), and the post-net — itself a universal approximator — builds the solution. The expressivity bottleneck is therefore the *post-net*, not the front end. Any front end spanning a comparable function space lets the post-net reach the same solution, which is precisely why the structurally-matched CFE (classical Linear + tanh front end) *and* a depth-matched classical net both tie the QAPINN (Section 5). The

config	model	params	rel-L_2 (mean $\pm$ std, n=3)	note
cv_cls	classical	121	$\mathbf{0.248 \pm 0.031}$	**best of all**
cv_qd5_lin	QAPINN d5	137	0.314 ± 0.073	depth helps; only ties CFE
cv_cfe	CFE control	109	0.315 ± 0.026	classical front-end
cv_qd3_none	QAPINN d3	121	0.369 ± 0.022	**no-ent beats linear**
cv_qd3_lin	QAPINN d3	121	0.424 ± 0.005	reference QAPINN (worst)

Table 5: Converged core (L-BFGS on, 3 seeds). Turning L-BFGS back on roughly halved every error vs. the undertrained runs (classical $0.535 \to 0.248$, CFE $0.474 \to 0.315$). At matched parameters the classical PINN beats every QAPINN.

quantum Fourier features are not a *better* basis for Burgers than classical features: away from the shock the solution is low-frequency and every basis represents it; at the shock it demands high harmonics that no basis supplies at depth 3. With no regime in which the Fourier/entangled basis is more efficient, there is no advantage to be had — which our fair controls confirm.

Why entanglement is neutral. **[reasoned]** At fixed depth the accessible band is fixed ($\{-L, \ldots, L\}$); entanglement only changes *which coefficients within that band* are reachable (the Mhiri constraint). But the post-net recombines the *unentangled* per-qubit Fourier features into the same target function, so that redistribution is largely redundant for this task. We tried to measure this (per-qubit feature correlation and effective rank, none vs. linear); the metrics conflicted (correlation rose with entanglement, but so did feature rank), so we report it as a theory-grounded interpretation consistent with the observed null, not a measured mechanism.

Why the encoding scale collapses the model at 2π. **[measured]** Scaling the encoding to $x \mapsto 2\pi x$ wraps a full period over the domain, so neighbouring collocation points map to widely separated circuit states (the realised spectrum aliases to the Nyquist frequency, verified). The loss surface around the true solution flattens and the optimiser descends into the trivial ($u \equiv 0$) basin that satisfies only the homogeneous BCs — the "black-hole" collapse, produced deterministically by an encoding choice.

Why "band = depth" holds for the features but not the model. **[measured]** The identity (max harmonic = re-upload depth) is a property of the VQC output, verified for $L = 1$–6. The post-net then applies tanh to combinations of these features; the Taylor expansion of tanh contains *products* of inputs, generating sum/difference frequencies far above the band. A converged depth-3 QAPINN thus realises harmonics to $k \approx 22$ (verified by FFT of the trained model), while an undertrained one stalls near $k \approx 3$. The band bounds the *features*; the nonlinear readout is unbounded.

Why the shock defeats every model. **[measured]** The Burgers shock is a near-discontinuity whose Fourier content decays only as $\sim 1/k$. By the F-principle (verified: low harmonics' error halves first) a gradient-trained network fits low frequencies early and the high-frequency shock content last — or not at all within budget. This is a *classical* PINN pathology; the quantum layer neither causes nor cures it, which is itself a mechanistic reason the quantum layer changes nothing where accuracy is decided.

Synthesis. The quantum component sits *upstream* of the part of the network that does the representational work. It contributes a fixed Fourier feature map whose useful content (a low-frequency

band) is equally available to a classical front end, and whose extra structure (entanglement, correlator choice) is redundant once a flexible post-net follows. The quantum layer's only decisive effects we observe are *negative* (the aliasing collapse) or *shared with the classical model* (spectral bias at the shock). Hence no quantum-specific advantage — for a reason, not merely as a measured null.

7 Discussion

Across two model scales, three control types, and multi-seed statistics, the honest reading is one-directional: *we find no statistically significant quantum-specific advantage for the QAPINN on 1-D Burgers.* At small scale the classical PINN simply wins; at resolved scale the QAPINN beats a shallow classical net but ties a depth-matched one, so the gain is architectural and classically reproducible. Entanglement shows no robust effect at either scale. The recurring pattern is the real lesson: *every apparent quantum advantage in this project dissolved under a fair control* — the "quantum diversifies the classical neurons" effect died under the CFE control; the "entanglement hurts" effect died under seed expansion; the "resolved-scale advantage" died under a depth-matched classical baseline. The value of the study is therefore methodological: a verified pipeline, matched, structural, and depth-matched controls, multi-seed statistics, and an adversarial self-audit that repeatedly caught our own over-claims — precisely the discipline the QAPINN literature usually omits, and without which each of those three over-claims would have been reported as a positive result.

The deliverable — a falsification protocol (our answer to Task 4). The challenge asks for a circuit-design methodology; our data do not warrant one (no robust advantage to design toward). What transfers instead is the four-gate protocol that killed our own over-claims: **(1)** train to convergence (L-BFGS), not Adam-only — most apparent gains were undertraining; **(2)** compare against a param-matched *and* structurally-matched *and* depth-matched classical control, not an arbitrary MLP; **(3)** require multi-seed significance ($n \geq 6$) — an $n=3$, $p=0.02$ effect here reversed at $n = 6$; **(4)** test Fourier-band claims on the VQC output, not the post-net-filtered model. An advantage that survives all four gates would justify a design methodology; ours survives none.

8 Limitations

The two converged findings (Section 5) are firm in direction but bounded in scope. (1) **Modest n and an underpowered null:** the converged cores use three-to-six seeds, and the decisive comparisons are *failures to reject* a difference (e.g. QAPINN-none vs. depth-matched classical, $p = 0.51$), under high seed variance (the 0.094 ± 0.075 QAPINN-none is $\sim$80% relative spread). More seeds would tighten these; we caught one $n = 3$ effect that reversed at $n = 6$, so we treat all single-scale nulls as provisional. (2) **Small-model regime:** at 121 parameters even the converged models sit at rel-L_2 0.25–0.42; the Burgers shock is better resolved than before but not fully resolved, and the orderings could shift with larger networks. (3) **Mechanism not yet isolated:** we have the number for a hypothetical entanglement effect that ultimately did not survive (§Finding 3). (4) **Pre-convergence ablations superseded:** the single-seed Tier-1/Tier-3 sections are undertrained and are retained only for transparency; their optimistic readings (bandwidth-gated advantage, barren-plateau signature) did not survive the audit. (5) **Simulation only** (`default.qubit`, statevector, no shot noise); parameter reduction is not resource reduction. A genuine quantum advantage at still-larger scale, more seeds, or on other PDEs is not excluded — only that we find

none here under fair controls. The pipeline is verified (Table 1) and every number matches its raw log.

9 Conclusion

Using a verified, parameter-matched bench, a five-dimension XAI suite, and an adversarial self-audit that repeatedly corrected our own over-claims, we reach a clear and honest result. Across two model scales, three control types (matched, structural, and depth-matched), and multi-seed statistics, **we find no statistically significant quantum-specific advantage** for the QAPINN on 1-D Burgers: at small scale the classical PINN wins; at resolved scale the QAPINN beats a shallow classical net but ties a depth-matched one, so the gain is architectural and classically reproducible; and **no robust entanglement effect** appears at either scale (an $n = 3$ "entanglement hurts" signal deflated at $n = 6$). The often-cited "VQC = Fourier band = depth" fact is verified but describes the quantum *features*, not the full model. The enduring contribution is the controlled methodology itself: three separate apparent advantages each dissolved under the appropriate fair control, and each would have been a false positive without it — which is precisely the discipline the QAPINN literature most needs. Whether a genuine advantage emerges at still-larger scale or on other PDEs is left open by, not contradicted by, this work.

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A Reproducibility

Every run is driven by a YAML config and a fixed seed; the full VQC configuration is serialised to `config.json`. Runs: `python run.py experiments/<name>.yaml`; ablations: `python run_ablation.py run`; correctness audit: `python verify_pipeline.py`. Simulator: PennyLane `default.qubit` (statevector, backprop). See `literature_review.md`, `methodology.md`, and `xai_methods.md` for the full method notes.